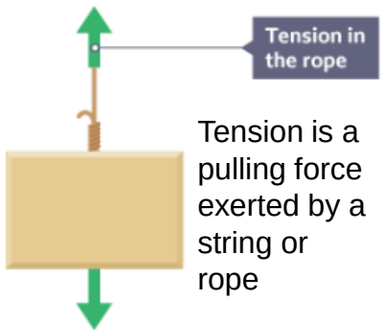
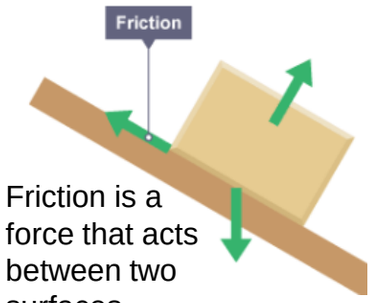
Knowledge Organiser
Year 7: Forces – Intro to Forces

What is a force?
- A force is a push or a pull that acts on an object due to an interaction with another object
- Forces are divided into **contact** forces and **non-contact** forces

These three diagrams show examples of **contact forces:**



Tension is a pulling force exerted by a string or rope



Friction is a force that acts between two surfaces

Air resistance happens when air particles hit a moving object

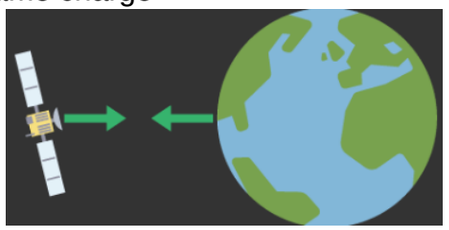
These three diagrams show examples of **non-contact forces:**



Magnetic attraction between two opposite poles of a magnet

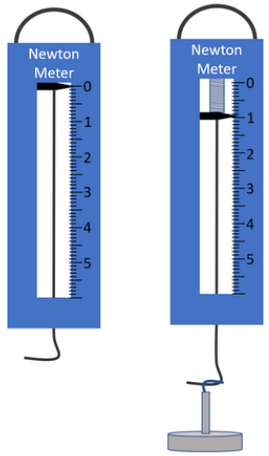


Electrostatic repulsion between two particles with the same charge



Gravitational attraction between two objects with mass

How do we measure forces?
- Forces can be measured using **a force meter**, also known as a Newton meter
- They have a spring inside them which moves a marker on a scale
- The further the spring is stretched, the bigger the force
- Forces are measured in **Newtons, N**.



A diagram showing a newton meter at zero and the same newton meter with a 1N weight attached to the hook.

What are mass and weight?
- Weight is an example of a non-contact force, it is measured in newtons, N
- It is caused by the pull of Gravity from Earth
- Mass is a measure of **amount of substance,** it is measured in grams, g
- It is linked to the number of **particles** an object has
- Larger masses have bigger weights



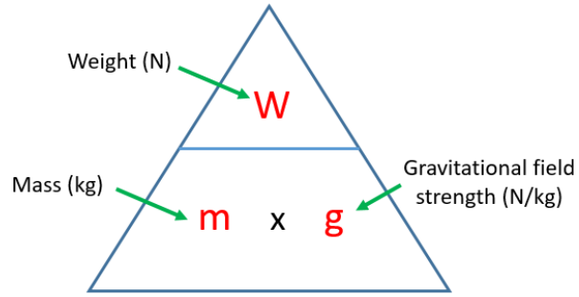


The moon has a smaller gravitational field strength than Earth.

Your weight would be lower on the moon, but your mass stays the same.

How do we calculate weight?
- Weight is the total amount of force on an object due to gravity
- The value of weight depends on the strength of the gravitational field
- On Earth, the gravitational field strength is 10N/kg

A formula triangle showing the equation which links **weight**, **mass** and **gravitational field strength**



Weight (N) = Mass (kg) x gravitational field strength (N/kg)

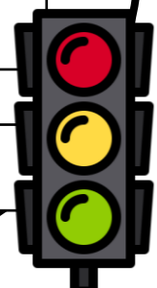
How do we draw forces?
- We can't see forces but can see their effects
- We draw forces as **arrows** on **force diagrams**
- The arrow shows the direction the force is acting
- The **size of the arrow** shows the size of the force



A force diagram showing the forces on a shopping cart being pushed at a constant speed

RAG Yourself!

Learning Outcomes;	RAG
Describe the difference between contact and non-contact forces	
State a range of contact and non-contact forces	
Draw forces arrows on an object	
Describe how to measure a force using a newton meter	
State the units of mass and weight	
Explain why weight changes on different planets but you mass does noy	



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